Quantum and Classical Portfolio Optimization: Continuous Allocation and Discrete Selection via QAOA

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Abstract

Portfolio managers routinely face a constraint that classical Markowitz optimization does not handle natively: a cap on the number of distinct positions a portfolio may hold. Selecting the best subset of B assets from a universe of N candidates is a combinatorial problem, and it is for this specific problem, not portfolio optimization in general, that quantum computing has been proposed as a candidate tool by both academic researchers and banks. This paper evaluates that proposal directly. Using an eleven-asset exchange-traded fund universe, we solve the continuous allocation problem exactly via classical convex optimization and the discrete selection problem using the Quantum Approximate Optimization Algorithm (QAOA), benchmarked against exact brute-force enumeration and classical simulated annealing. Across five tested position limits, QAOA recovered the true optimal portfolio every time, matching classical methods exactly, but it did so without confidently distinguishing the right answer from worse ones in its output distribution. We argue this result is informative for financial practice: quantum methods may eventually assist with cardinality-constrained selection at scale, but the technology is not yet a reliable substitute for classical heuristics, and claims to the contrary should be treated with skepticism.

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1. Introduction

Portfolio optimization is the problem of deciding how to divide capital among a set of available assets so as to achieve the best attainable trade-off between expected return and risk. Since Harry Markowitz's original formulation in 1952, this problem has occupied a central place in both academic finance and investment practice, because it converts a vague goal, build a good portfolio, into a precise mathematical question with a computable answer. In its classical form, the problem is well understood and computationally easy: given a set of assets with known (or estimated) expected returns and a covariance structure describing how their returns move together, a portfolio manager can solve for the exact mix of holdings that minimizes risk for any target level of return. This is the version of portfolio optimization taught in every introductory finance course, and it remains the analytical backbone of index construction, asset allocation, and risk budgeting across the industry.

In practice, however, portfolio managers rarely face the unconstrained version of this problem. A fund may be limited by mandate to holding no more than a fixed number of distinct positions; a high-frequency strategy may need to avoid the transaction costs and operational overhead of monitoring dozens of small holdings; a retail-facing product may want a concentrated, easily explainable set of names rather than a diffuse basket. Each of these situations imposes a cardinality constraint: out of N available assets, the portfolio may hold at most, or exactly, B of them. This single restriction changes the mathematical character of the problem entirely. Whereas the unconstrained problem is convex and solvable in closed form, the cardinality-constrained problem requires deciding which specific assets to include, a combinatorial decision with no continuous relaxation, and the problem becomes NP-hard: the time required to guarantee an exact answer grows explosively with the number of candidate assets, and no algorithm is known that solves it efficiently at scale (Chang et al., 2000; Bienstock, 1996; Zhang et al., 2018).

It is precisely this combinatorial bottleneck, not portfolio optimization in general, that has drawn the attention of quantum computing researchers and a growing number of financial institutions. Quantum computers are, in principle, well suited to certain combinatorial search problems because they can represent and manipulate a superposition over many candidate solutions simultaneously, offering a plausible path to exploring a large solution space more effectively than a classical computer checking candidates one at a time. This paper investigates that claim on a problem small enough to check the answer by brute force, whilst remaining structurally identical to the much larger cardinality-constrained problems that arise in real portfolio management. We solve the continuous allocation problem exactly using standard convex optimization, and we solve the discrete selection problem using the Quantum Approximate Optimization Algorithm (QAOA), a near-term quantum algorithm that has been piloted by financial institutions including Citigroup, in partnership with the quantum software firm Classiq, specifically for this use case (McKinsey & Company, 2025).

The paper is thus organized as follows. Section 2 reviews the relevant literature: on cardinality-constrained portfolio selection in classical finance and on the use of quantum algorithms in financial optimization, to establish both the practical motivation for this study and the specific gap it addresses. Section 3 develops the classical Markowitz baseline. Section 4 formulates the discrete selection problem mathematically required for quantum computation. Section 5 describes the classical benchmarks used to evaluate quality. Section 6 will lay out the QAOA implementation. Section 7 presents results across five position limits. Section 8 discusses the financial implications of these results and their limitations. Section 9 concludes.

2. Literature Review

2.1 Cardinality Constraints in Classical Portfolio Theory

The difficulty of cardinality-constrained portfolio selection has been recognized in the finance and operations research literature for decades. Chang, Meade, Beasley, and Sharpe (2000) were among the first to formalize the cardinality-constrained extension of the Markowitz model as a mixed-integer quadratic program and to demonstrate, using genetic algorithms and other heuristics, that exact solutions become impractical to guarantee as the number of candidate assets grows. Bienstock (1996) independently established the NP-hardness of the problem from an operations-research perspective, confirming that the difficulty is not an artifact of any particular solution method but a structural feature of the problem itself. More recent work has continued to treat the cardinality constraint as one of the central “real-world” frictions that separates textbook Markowitz theory from investable portfolio construction, alongside transaction costs and minimum trade sizes (Zhang, Leung, & Aravkin, 2018; Cesarone, Scozzari, & Tardella, 2013).

This matters for practitioners for an unglamorous but concrete reason: every additional position in a portfolio carries a fixed cost, in trading commissions, custody, compliance monitoring, and the analyst time required to track the position, that a purely continuous optimization ignores. A fund mandate that caps the number of names a manager may hold is therefore not an arbitrary inconvenience but a direct encoding of these costs, and solving the resulting selection problem well, choosing which handful of assets is truly worth holding, has real economic value distinct from the value of continuous allocation. Classical approaches to this problem, including genetic algorithms, simulated annealing, and continuous relaxations of the integer program, remain effective at the asset-universe sizes typically considered in academic studies, but none offers a guarantee of finding the true optimum once the universe grows into the hundreds or thousands of candidate assets that a large institutional manager might screen.

2.2 Quantum Computing in Finance Problems

Whilst the main charm of quantum computing in many spaces is that it can manage large-scale computations with greater efficiency, there is also a more technical chain of reasoning for how it may serve financial optimization in particular. A quantum computer represents information using qubits, which, unlike classical bits, can exist in superposition, and a circuit of suitably entangled qubits can be used to represent a weighted combination of many candidate solutions to a combinatorial problem at once. For problems like cardinality-constrained portfolio selection, where the feasible region is a large but finite set of discrete subsets rather than a continuous surface, this property is theoretically attractive: it offers a path, at least in principle, to searching the space of candidate subsets more efficiently than checking them one at a time. Orús, Mugel, and Lizaso (2019) provide an early and widely cited survey making exactly this case, framing portfolio optimization as one of the most natural financial applications of near-term quantum algorithms precisely because of its combinatorial structure. Herman et al. (2022) and subsequent systematic reviews have since catalogued a substantial and growing body of work applying quantum and quantum-inspired methods to this and closely related financial optimization problems (Slate, Matwiejew, Marsh, & Wang, 2021; Kerenidis, Prakash, & Szilágyi, 2019).

The Quantum Approximate Optimization Algorithm, the specific method used in this paper, is one of the most extensively studied candidates within this literature, mostly because it is designed for exactly the hardware available today. Current quantum computers operate in what is generally called the noisy intermediate-scale quantum, or NISQ, regime; they have enough qubits to represent interesting problems but not enough error correction to run deep computations reliably. QAOA is therefore a better choice in this setting because it uses comparatively short quantum circuits, with most of the heavy computational lifting handled by an ordinary classical computer in a feedback loop, making it one of the more hardware-realistic proposals for near-term quantum advantage in combinatorial optimization generally and portfolio selection specifically (Farhi, Goldstone, & Gutmann, 2014; Egger et al., 2022). Several direct benchmarking studies of QAOA on portfolio-selection problems already exist in the physics and quantum-computing literature, including detailed analyses of how circuit noise and finite measurement sampling degrade solution quality relative to the noiseless ideal (Slate et al., 2021), and comparisons between QAOA and alternative variational quantum algorithms on the same class of problem (Liu et al., 2022).

Financial institutions have taken this proposal seriously enough to test it directly rather than treat it as purely academic. Citigroup partnered with the quantum software company Classiq to pilot QAOA specifically for portfolio optimization, examining in particular how the algorithm's constraint-penalty parameter affects solution quality, the same parameter this paper refers to as λ (McKinsey & Company, 2025). JPMorgan Chase has maintained a dedicated quantum research team of more than fifty scientists investigating portfolio construction and related use cases, and has published technical work on quantum algorithms for financial optimization (Pistoia, cited in American Banker, 2024). At the same time, the most prominent public account of an internal feasibility study, conducted at Goldman Sachs, concluded that a fault-tolerant quantum algorithm capable of meaningfully outperforming classical portfolio optimization at realistic scale would require on the order of millions of error-corrected logical qubits, a hardware milestone that remains, by a wide margin, beyond any presently announced quantum computing roadmap (Bloomberg, 2026). This divergence between institutions exploring the same underlying technology is itself informativ, as it reflects a genuine, unresolved disagreement within the financial industry about how close quantum computing is to delivering practical value for problems of exactly this kind.

This paper situates itself within that disagreement empirically rather than rhetorically. Rather than argue in the abstract about whether quantum computing will eventually help with portfolio construction, we implement the specific algorithm at the center of the industry's pilots, on a problem small enough to check the answer exactly, and report what it actually does: whether it finds the right answer, how confidently, and under what conditions it does not. This contributes to the literature in a direction distinct from most existing QAOA-portfolio studies, which tend to report final solution quality without always making explicit, for the financial world, the distinction between locating an optimal solution within a sampled distribution and confidently identifying it, a distinction with direct consequences for whether a result of this kind could actually be trusted in a production investment process.

3. Data and the Classical Markowitz Baseline

3.1 Data Overview

The asset universe considered throughout this study comprises eleven exchange-traded funds, each tracking a distinct sector of the U.S. equity market or a distinct asset class: financials (XLF), energy (XLE), technology (XLK), healthcare (XLV), industrials (XLI), consumer discretionary (XLY), consumer staples (XLP), utilities (XLU), gold (GLD), long-term U.S. Treasuries (TLT), and real estate investment trusts (VNQ). This selection spans equity sectors with meaningfully different risk and growth characteristics alongside two non-equity asset classes, gold and long-duration government debt, whose returns are frequently uncorrelated or negatively correlated with equities and which therefore play a diversifying role in any mean-variance analysis.

Daily adjusted closing prices were obtained for all eleven tickers over a trailing three-year window, yielding 757 trading days of fully overlapping data. Three years balances two competing concerns: a shorter window would leave too few observations to estimate the fifty-five free parameters of an eleven-by-eleven covariance matrix with much statistical confidence, while a substantially longer window risks blending in stale, regime-shifted data, for instance from the 2022 monetary tightening cycle, whose volatility and correlation patterns may no longer describe current markets well. Daily log returns were computed from the price series according to:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right), \qquad \mu = 252 \cdot \overline{r}, \qquad \Sigma = 252 \cdot \mathrm{Cov}(r)$$

where P_t denotes the adjusted closing price on day t and the factor of 252 annualizes using the conventional number of trading days per year. The resulting expected-return vector, μ, and covariance matrix, Σ, are the only inputs required for both the continuous and discrete sides of this study.

A rolling-window check was used to assess how stable this covariance estimate is over time, computing Σ on successive overlapping one-year windows and tracking how much it shifts between windows. The typical shift was moderate, but the largest single shift was substantial, consistent with a volatility regime change somewhere within the three-year sample. This is reported here as a limitation rather than smoothed over: any static covariance matrix is an approximation to a true relationship between assets that drifts over time, and a portfolio manager relying on a single historical estimate should bear that drift in mind, even though three years remains a reasonable, standard window length given the trade-off against noisier estimates at shorter horizons.

Table 1 summarizes the resulting annualized return and volatility for each asset. Technology and gold delivered the highest annualized returns over the sample period, while long-term Treasuries delivered a small negative return, consistent with the well-documented multi-year drawdown in long-duration government debt as policy rates rose from their post-2020 lows. Utilities, consumer staples, and healthcare exhibit the lowest volatility, consistent with their conventional classification as defensive sectors.

Table 1. Annualized expected return and volatility by asset, three-year trailing sample.

Ticker	Sector / Asset Class	Annualized Return (μ)	Annualized Volatility (σ)
XLF	Financials	0.1742	0.1604
XLE	Energy	0.1338	0.2172
XLK	Technology	0.2561	0.2424
XLV	Healthcare	0.0650	0.1403
XLI	Industrials	0.1900	0.1637
XLY	Consumer Discretionary	0.1126	0.2079
XLP	Consumer Staples	0.0681	0.1240
XLU	Utilities	0.1319	0.1643
GLD	Gold	0.2432	0.2048
TLT	Long-Term Treasuries	-0.0180	0.1388
VNQ	REITs	0.0914	0.1687

3.2 The Continuous Mean-Variance Problem

The classical long-only Markowitz problem seeks a weight vector w, with one entry per asset, that minimizes portfolio variance subject to a minimum acceptable expected return and the requirement that capital be fully invested without short positions. Formally:

$$\min_{w} w^{\top}\Sigma w \quad \text{s.t.} \quad \mu^{\top} w \geq r^{*}, \;\; 1^{\top} w = 1, \;\; w \geq 0$$

where r* denotes the minimum acceptable expected return for a given point on the efficient frontier. This is a convex quadratic program, which guarantees that any solution found is the true global optimum, not merely a good local one, and that it can be found quickly regardless of how many assets are in the universe. The efficient frontier is traced out by solving this program repeatedly across a range of target returns, from the return of the minimum-variance portfolio up to the return of the single best-performing asset.

A particular point on the frontier of practical interest is the portfolio that maximizes the Sharpe ratio, the ratio of excess return to volatility most commonly used to compare risk-adjusted performance across strategies. This portfolio can be found exactly, without a grid search across the frontier, using a change-of-variables technique due to Cornuejols and Tutuncu: defining a rescaled variable y such that w is recovered by renormalizing y to sum to one, the Sharpe-maximization problem becomes a second convex quadratic program with a single linear constraint:

$$\min_{y} y^{\top}\Sigma y \quad \text{s.t.} \quad (\mu - r_f)^{\top} y = 1, \;\; y \geq 0; \qquad w = \frac{y}{1^{\top} y}$$

after which the maximum-Sharpe weights are recovered as w = y / (1^{t}y). Both quadratic programs were solved using the cvxpy convex optimization package in well under a second; this is the entire reason no quantum or other unconventional method is applied to this part of the problem, since classical convex optimization already solves it exactly and essentially instantly.

3.3 Classical Results

The global minimum-variance portfolio achieves an annualized expected return of 0.0830 at an annualized volatility of 0.0884, for a Sharpe ratio of 0.939. It concentrates capital in consumer staples (0.3099) and long-term Treasuries (0.3007), with smaller allocations to gold (0.1132), healthcare (0.0834), energy (0.0902), technology (0.0543), and financials (0.0484). This is intuitive: the minimum-variance solution favors the lowest-volatility sectors and an asset class, long Treasuries, that has historically diversified against equity risk, even though Treasuries happened to underperform over this particular three-year sample.

The maximum-Sharpe portfolio achieves a substantially higher annualized return of 0.2023 at a volatility of 0.1233, for a Sharpe ratio of 1.641. It concentrates capital in gold (0.3694) and financials (0.2953), with secondary allocations to technology (0.1606) and utilities (0.0940). Notably, the single highest-return asset in the universe, technology, receives a meaningful but not dominant weight once its volatility and correlation with the rest of the portfolio are accounted for, which is the ordinary and expected behavior of mean-variance optimization: the best asset in isolation is not necessarily the best asset to be overweight in a diversified, risk-adjusted portfolio.

The complete efficient frontier, spanning fifty points from the minimum-variance return up to just below the best single-asset return, is shown in Figure 1 alongside both named portfolios and the risk-return position of every individual asset.

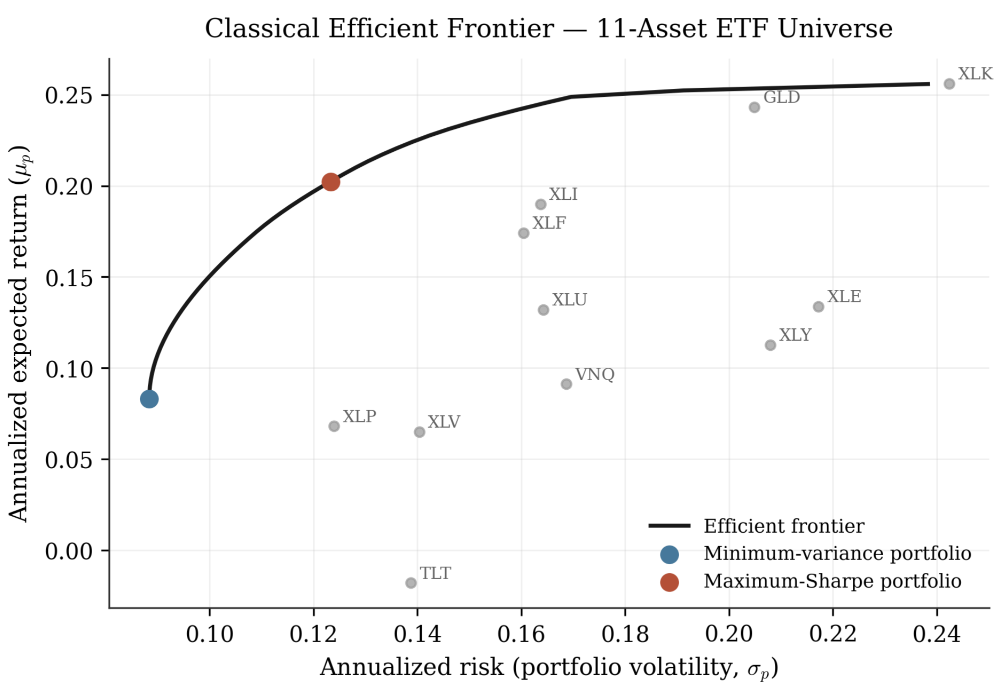

Figure 1. The classical efficient frontier for the eleven-asset universe, computed via convex quadratic programming. The minimum-variance portfolio (blue) and maximum-Sharpe portfolio (red) are marked; individual assets are shown in gray for reference.

4. From the Discrete Selection Problem to Quantum Computation

4.1 Overview

Section 3 answered the question of how to divide capital across all eleven assets at once. A different question, the one motivating this paper, arises when an investor is limited to holding only B of the eleven assets, with no partial credit for assets outside the chosen subset: which B assets jointly deliver the best risk-adjusted outcome? As discussed in Section 2.1, this is not a continuous optimization problem; it is a search over (11 choose B) discrete candidate subsets, and that search is NP-hard in general, meaning no algorithm is known that guarantees the exact best answer in a time that scales gently as the universe of candidate assets grows. This is the problem for which this paper builds and tests a quantum computational approach.

To make this problem solvable on a quantum circuit, it must first be rewritten in a specific mathematical form called a Quadratic Unconstrained Binary Optimization problem, or QUBO. The idea is straightforward: every asset is assigned a single yes-or-no variable indicating whether it is held, and the entire objective, both the quantity to be minimized and the limit on how many assets may be selected, is folded into one quadratic expression in those yes-or-no variables. The word “unconstrained” in the name reflects a real limitation of the format: a QUBO objective cannot enforce a rule directly, so the requirement that exactly B assets be selected has to be built into the objective itself as a penalty that makes violating it costly. Writing w_i as the binary variable indicating whether asset i is held, the QUBO objective used in this study is:

$$H(w) = w^\top \Sigma w \; - \; q\mu^\top w \; + \; \lambda\left(\sum_{i=1}^{n} w_i - B\right)^2, \qquad w_i \in \{0,1\}$$

The three terms each serve a distinct financial purpose. The first term, w-transpose Σ w, is the portfolio variance contributed by the selected assets; because Σ captures how every pair of assets co-moves, this term penalizes selections of mutually correlated assets and rewards selections that diversify, exactly as the variance term does in ordinary Markowitz optimization. The second term, minus q times μ-transpose w, rewards selecting assets with high expected return, with q acting as a risk-aversion dial analogous to the trade-off parameter implicit at any point on the classical efficient frontier; this study fixes q at one, weighting risk and return equally. The third term is the penalty enforcing the position limit: it squares the gap between the number of assets actually selected and the target B, so that selecting too many or too few assets becomes costly in proportion to the square of that gap.

Expanding this penalty term algebraically, and using the fact that any binary variable satisfies w_i squared equals w_i, shows that it simply adds a fixed amount to each diagonal entry of the QUBO coefficient matrix and another fixed amount to every off-diagonal entry:

$$Q_{ii} \leftarrow Q_{ii} + \lambda(1-2B), \qquad Q_{ij} \leftarrow Q_{ij} + \lambda \;\; (i \neq j)$$

Getting the size of this penalty right, denoted λ, matters more than any other modeling choice in this paper, and it is worth being direct about why. A penalty that is too weak can let the optimizer get away with selecting the wrong number of assets, if doing so sufficiently improves the risk and return terms; a penalty that is too strong drowns out the risk and return information entirely, so that the position limit is satisfied but the resulting selection says nothing meaningful about portfolio quality. This study sets λ using a conservative scaling rule based on the largest plausible swing in the risk and return terms from changing a single asset's inclusion, and then checks the choice empirically: for every position limit tested, the unconstrained optimum of the penalized objective coincided exactly with the best selection that actually satisfies the position limit, found by exhaustive search. Had λ been poorly chosen, these two would have disagreed. This is reassuring evidence that the penalty was well calibrated for this specific problem, though it is not a guarantee that the same scaling would work unchanged for a different universe of assets or a materially different position limit.

4.2 Mapping to a Quantum Circuit

A quantum computer cannot act on a QUBO matrix directly; it operates on qubits, whose underlying physics is naturally described in the language of quantum spins rather than zeros and ones. The QUBO objective is therefore translated one more time, into what is called an Ising Hamiltonian, the mathematical object that actually gets encoded onto a quantum circuit. This translation substitutes each binary variable with a spin variable taking the value plus or minus one, after which the objective takes the general form of a sum of single-qubit terms plus a sum of pairwise-coupling terms between qubits. Because the original covariance matrix links every pair of the eleven assets, with no covariance assumed to be exactly zero, the resulting problem is fully connected: every one of the eleven qubits is coupled to every other qubit, sixty-six couplings in total. This is a meaningful practical detail, not just a mathematical curiosity, because a quantum circuit solving a fully connected problem needs two-qubit operations between every qubit pair, a requirement that grows quickly with the number of assets and is one of the concrete, hardware-level reasons today's quantum computers struggle to scale this kind of problem much beyond the size considered in this paper. This conversion was performed programmatically using Qiskit's built-in tools, ensuring it was applied identically and without manual error across all five position limits tested.

5. Classical Reference Solvers

Before any claim can be made about how well a quantum algorithm performs, its output has to be checked against a trustworthy standard. This study uses two classical reference methods for that purpose: exact brute-force enumeration, which remains fully tractable at this problem size and therefore establishes unambiguous ground truth, and classical simulated annealing, a widely used heuristic whose comparison to QAOA is the comparison that would actually matter at the larger problem sizes found in real institutional portfolios, where brute-force checking is no longer an option.

5.1 Brute-Force Exact Enumeration

With eleven assets, the total number of possible selections is:

$$2^{11} = 2048 \ \text{ total bitstrings, fully enumerable}$$

a number small enough to check exhaustively on an ordinary laptop in a fraction of a second. For each of the five position limits tested, every one of the 2,048 possible selections was scored under the QUBO objective, and the best-scoring selection that actually satisfied the position limit was recorded as the true optimum for that limit. This same exhaustive check, as discussed in Section 4.1, was also used to validate the penalty calibration. Brute-force enumeration completed in under ten milliseconds for every position limit, underscoring that the eleven-asset universe used in this study, while large enough to expose genuine and informative QAOA behavior, is far too small to itself justify a quantum computational approach on practical grounds; the value of the exercise lies in establishing a benchmarking methodology that extends cleanly to the much larger, classically intractable universes that real institutional portfolios actually draw from.

5.2 Classical Simulated Annealing

Simulated annealing, a standard optimization heuristic that searches the space of candidate selections while gradually becoming less willing to accept worse solutions over time, was run with twenty independent restarts per position limit, each proceeding for two thousand iterations with a cooling schedule decreasing from an initial temperature of ten to a final temperature of 0.01. Simulated annealing recovered the exact brute-force optimum at every position limit tested, in well under half a second of computing time per limit. This agreement is a useful result in its own right: it confirms that a comparatively simple, decades-old classical method is entirely sufficient to solve this particular eleven-asset problem, and it sets the bar that any quantum method would need to clear to offer real practical value once the universe grows too large for brute-force checking.

6. The Quantum Approximate Optimization Algorithm

6.1 Overview of QAOA

QAOA works by repeatedly running a quantum circuit, checking on average how good the resulting answer is, and letting an ordinary classical computer adjust the circuit's settings before trying again. The quantum and classical components hand off to each other in a loop until the settings stop improving. The quantum part of this loop is what makes QAOA worth testing at all: the circuit prepares a state that represents a weighted combination of many candidate asset selections simultaneously, rather than considering one selection at a time the way a classical search does, and the hope, grounded in the surveyed literature in Section 2.2 rather than asserted here, is that this structure lets the algorithm explore the combinatorial space of selections more efficiently than classical enumeration once the number of assets grows large.

QAOA is specifically suited to present-day hardware because it uses short quantum circuits and pushes most of the computational burden onto a classical optimizer running outside the quantum device, which makes it far more tolerant of the noise and limited qubit counts that characterize current, pre-error-corrected quantum computers than algorithms requiring long, deep circuits. This is precisely why QAOA, rather than a more theoretically powerful but currently unrealizable algorithm, is the method actually being piloted by financial institutions today, and why it is the method tested in this paper. More formally, the algorithm prepares a parameterized quantum state by alternately applying, for a fixed number of layers p, an evolution under the problem Hamiltonian H_C derived in Section 4.2 and an evolution under a fixed mixing Hamiltonian H_M, starting from an equal superposition over all possible selections. Formally, the prepared state is:

$$|\psi(\boldsymbol{\beta},\boldsymbol{\gamma})\rangle = \prod_{l=1}^{p} e^{-i\beta_l H_M} e^{-i\gamma_l H_C} \, |+\rangle^{\otimes n}$$

where β and γ are vectors of p real-valued settings, one pair per layer, and the mixing Hamiltonian is conventionally the sum of single-qubit Pauli-X operators:

$$H_M = \sum_{i=1}^{n} X_i$$

whose role is to keep the algorithm exploring the full space of selections rather than settling prematurely near its starting point. The best settings are, in principle, those minimizing the expected value of the problem Hamiltonian under the prepared state:

$$(\boldsymbol{\beta}^*,\boldsymbol{\gamma}^*) = \arg\min_{\boldsymbol{\beta},\boldsymbol{\gamma}} \langle\psi(\boldsymbol{\beta},\boldsymbol{\gamma})| H_C \,|\psi(\boldsymbol{\beta},\boldsymbol{\gamma})\rangle$$

a quantity that cannot be computed directly except for the smallest problems and must instead be estimated by running the circuit repeatedly and measuring the outcomes, with that estimate handed to a classical optimizer that proposes new settings, in a loop that continues until the settings converge or a fixed budget of attempts is used up.

6.2 Implementation and a Documented Software Incompatibility

This study's implementation departs in one respect from the most commonly cited Qiskit tutorial pattern, and the departure is documented here because it reflects a genuine, currently unresolved versioning issue within the Qiskit software ecosystem rather than a deliberate choice. The qiskit_algorithms package offers a convenience class that wraps the entire QAOA loop behind a single interface, but that class was written against an older generation of Qiskit's execution interface and is incompatible with the current release used throughout this study; attempting to use it produced a failure before any quantum circuit execution actually occurred. Rather than downgrade the software stack to chase compatibility with a deprecated interface, this study implements the QAOA loop directly using Qiskit's modern circuit-construction and execution tools, with the COBYLA optimization routine from SciPy serving as the classical outer loop. This has the added benefit of making every component of the loop, the circuit, the cost being minimized, and the optimizer's convergence behavior, fully visible and checkable rather than hidden behind one method call.

For each tested position limit, a two-layer QAOA circuit was built from the Ising Hamiltonian derived in Section 4.2, its settings initialized to small random values, and then optimized for up to two hundred iterations using COBYLA, with each iteration estimating the expected cost via repeated circuit measurement on a noiseless quantum simulator. Once the settings converged, the final circuit was sampled 4,096 times, producing a full distribution over the 2,048 possible asset selections rather than a single answer.

QAOA's optimization landscape, like that of most non-convex problems, has multiple local optima, and the quality of the settings found by any single optimization run depends heavily on its starting point, which is arbitrary. Reporting only a single run's result would risk making QAOA look either better or worse than it typically performs, depending on luck. This study instead follows standard practice in the quantum algorithms literature and uses a multi-start protocol: five independent runs from five different random starting points were performed for each position limit, and the run achieving the best converged cost was treated as that limit's representative result, with the remaining four retained to report how much variability exists across starting points.

7. Results

7.1 Summary Across Position Limits

Table 2 reports the full comparison across all five tested position limits, B equal to three through seven assets out of eleven. For each limit, the table shows the brute-force optimal selection and its equal-weight Sharpe ratio, whether simulated annealing matched that optimum, whether QAOA's best-of-five-restarts run matched it, the resulting energy gap, and the share of QAOA's 4,096 measurements that satisfied the position limit at all, averaged across the five restarts.

Table 2. Full benchmark results across five position limits.

B	Optimal Selection	Optimal Sharpe	SA Matches Optimum	QAOA Matches Optimum	QAOA Energy Gap	QAOA Mean Feasible-Shot %
3	XLF, XLK, GLD	1.564	Yes	Yes	0.000	22.7%
4	XLF, XLK, XLU, GLD	1.584	Yes	Yes	0.000	30.5%
5	XLF, XLK, XLI, XLU, GLD	1.548	Yes	Yes	0.000	61.5%
6	XLF, XLK, XLI, XLP, XLU, GLD	1.528	Yes	Yes	0.000	44.1%

7	XLF, XLE, XLK, XLI, XLP, XLU, GLD	1.486	Yes	Yes	0.000	27.7%

The selected portfolios show a sensible nested structure as the position limit relaxes: financials, technology, and gold appear in the optimal selection at every limit from three through seven, indicating that these three assets, by virtue of their favorable combination of return, volatility, and mutual diversification, are attractive enough to be retained regardless of how many additional positions are allowed. As the limit relaxes further, additional assets are progressively admitted, utilities at four positions, industrials at five, consumer staples at six, and energy at seven, each addition reflecting the next-best marginal contribution once the core three-asset combination is already in place. Figure 2 visualizes this pattern directly.

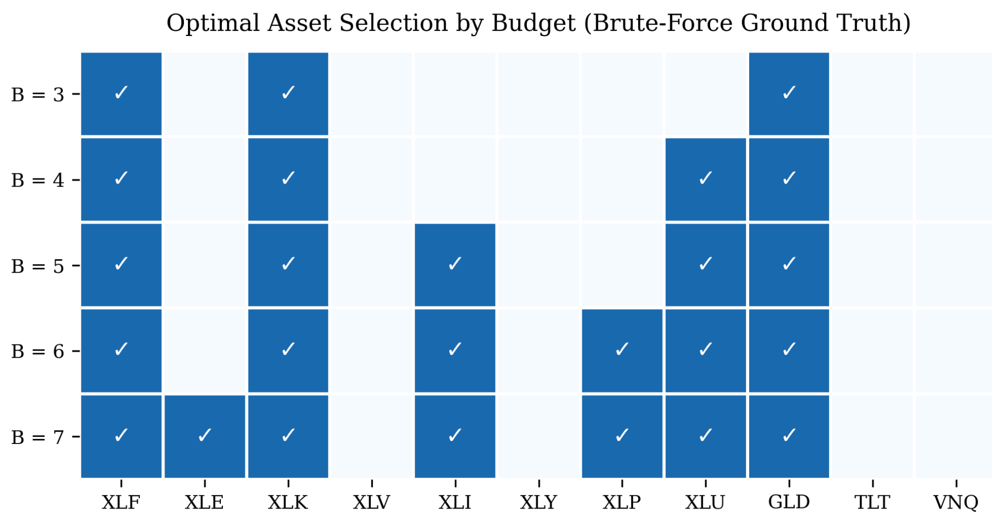

Figure 2. Brute-force optimal asset selection at each tested position limit, B equal to three through seven. Financials (XLF), technology (XLK), and gold (GLD) are selected at every limit; additional assets enter in order of marginal contribution as the limit relaxes.

The equal-weight Sharpe ratio of the optimal selection also decreases mildly and consistently as the position limit relaxes, from 1.564 at three positions down to 1.486 at seven. This is expected under equal weighting. As additional, individually less attractive assets are forced into the portfolio by a looser limit, the equal-weight average is diluted by assets that a continuous reallocation would assign a smaller share. This reinforces the separation between the two problems studied in this paper: a continuous reallocation within any fixed selected subset would likely improve on the equal-weight Sharpe ratio reported here, and the equal-weighting convention is acknowledged in Section 8 as a simplification, not a claim about how a manager would actually size positions once a subset has been chosen.

7.2 QAOA Performance

The central finding of this study is that QAOA's best-of-five-restarts run located the exact brute-force optimal portfolio at every one of the five tested position limits, with zero gap in every case. This is a genuine and non-trivial result for an eleven-qubit, fully connected problem addressed with only two circuit layers, and it should be taken seriously as evidence that QAOA, in principle, can find the right answer to this style of problem.

However, locating the right answer within a sampled distribution is a substantially weaker claim than identifying it with any confidence, and the gap between these two claims is the most financially important finding in this study. Across the five position limits, the share of all measurements satisfying the position limit at all, regardless of whether the resulting selection was optimal, ranged from 22.7 percent at three positions up to 61.5 percent at five positions, with no clean relationship to the position limit itself; six and seven positions, despite allowing more valid selections in absolute terms, did not produce correspondingly higher shares. Figure 3 displays this pattern.

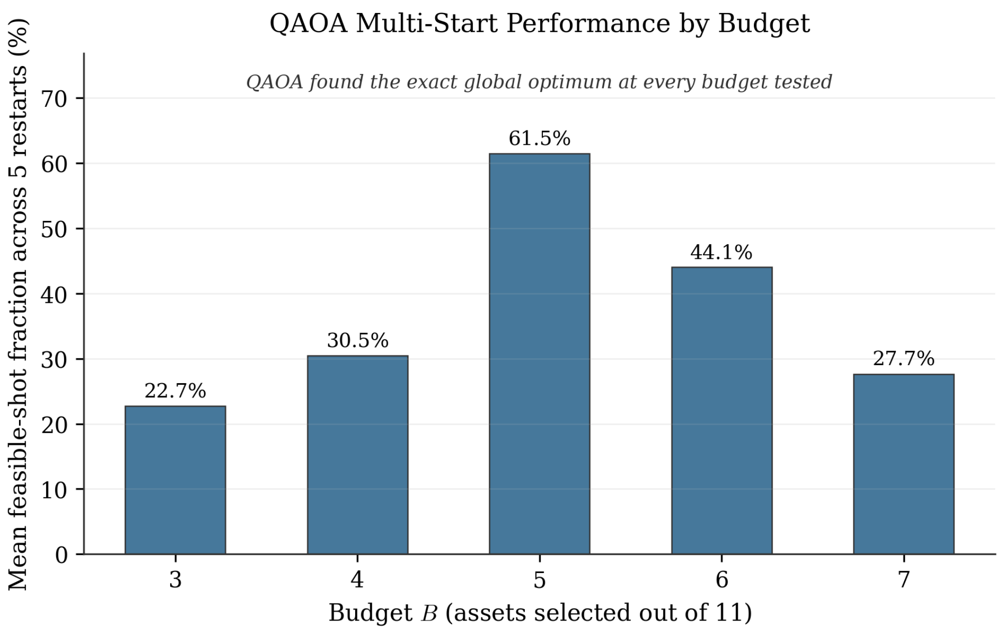

Figure 3. Mean share of QAOA measurements satisfying the position limit, averaged across five random restarts, by limit. The absence of a clean trend across limits indicates that this share is governed by more than limit size alone, most plausibly the interaction between the penalty term and how well the classical optimizer converges at each specific setting.

Within even the best-converged run at any given limit, the probability the final measurement distribution placed on the single optimal selection was typically a small fraction of one percent. The single most frequently measured selection, in several cases, was not the optimal one at all, and in early single-run exploration before the multi-start protocol was adopted, the most frequent outcome was sometimes the selection of no assets whatsoever, a result that violates the position limit outright. This is a known weakness of shallow QAOA circuits applied to penalty-based objectives of this kind: the trivial all-or-nothing selections often sit in a smooth, easy-to-reach region of the optimization landscape near the algorithm's starting point, and a classical optimizer working with a limited number of attempts can settle on settings that leave meaningful probability on these trivial, financially useless outcomes even when a much better selection exists elsewhere in the same distribution.

A supplementary check at the four-position limit examined whether adding more circuit layers, from one up to three, improved the share of valid measurements at a fixed classical-optimizer budget of two hundred attempts. The result did not improve steadily: the valid share rose from 9.0 percent at one layer to 39.5 percent at two layers, then fell back to 11.3 percent at three layers, even though the optimal selection appeared in the measured distribution at every depth. This suggests that, at least within the attempt budget tested, the main constraint on QAOA's reliability here is not circuit depth by itself but the interplay between circuit depth and how well the classical optimizer can search an increasingly complex settings landscape within a fixed number of attempts.

8. Discussion

8.1 Financial Implications

The practical question this paper set out to answer was whether QAOA, the specific quantum algorithm currently being piloted by institutions such as Citigroup, can be trusted today to solve cardinality-constrained portfolio selection in place of, or alongside, classical heuristics. The honest answer, based on the evidence in Section 7, is not yet, and the reason is precise rather than vague: QAOA reliably contains the right answer somewhere within its output, but it does not reliably tell the user which output is the right one. A portfolio manager using this algorithm today, drawing a single sample or even a modest number of samples from the circuit's output, would have no principled way to distinguish the optimal selection from a mediocre or outright invalid one without already knowing the answer by some other means, which defeats the purpose of running the algorithm in the first place. This is a meaningfully different and more specific conclusion than simply stating that quantum computers are not yet powerful enough, a claim repeated often enough in industry commentary to have lost much of its precision; the specific failure mode here is a confidence problem, not a search problem, since the search half of the algorithm's job was accomplished successfully at every position limit tested.

This finding is consistent with, and helps explain in concrete terms, the caution expressed by Goldman Sachs in scaling back its own quantum portfolio research, whilst also helping explain why JPMorgan Chase and Citigroup continue to invest in exploring the same technology: the algorithm is not obviously useless, since it does find correct answers, but it is not yet reliable enough to deploy in a process where a manager cannot independently verify the result. For a financial institution deciding how much to invest in this technology today, the relevant takeaway is not a solid verdict on whether quantum computing works, but a more specific diagnosis of where the current bottleneck lies: in extracting a confident answer from a correct but noisy distribution, a problem that may be addressed in the future through better classical optimizers paired with the quantum circuit, deeper circuits once hardware error rates fall, or alternative variational algorithms such as Layer-VQE, which has shown more favorable scaling than QAOA in some published comparisons (Liu et al., 2022). None of these fixes are available today at the scale a real institutional portfolio would require.

The future directions for this paper are thus also evident. The benchmarking methodology developed in this paper, comparing a quantum algorithm against both an exact classical answer and a realistic classical heuristic, rather than against an idealized theoretical bound, is directly reusable at the much larger asset universe sizes where classical brute-force checking stops being possible and where a genuine practical case for quantum methods could eventually emerge. The eleven-asset universe tested here was chosen specifically because it is large enough to expose real algorithmic behavior, including the confidence problem just described, while still being small enough to verify every claim against ground truth. Extending this same methodology to a fifty- or hundred-asset universe, where brute-force verification is no longer an option, is the natural next step, and the present results should be read as establishing what to look for when that extension is undertaken: not just whether the algorithm finds a good answer, but how confidently it does so.

8.2 Limitations

We acknowledge several limitations of this paper. The penalty coefficient λ governing the position limit was set using a heuristic rule and validated only empirically, by confirming after the fact that it produced consistent results at every tested limit; this is reassuring for this specific problem but is not a theoretical guarantee that the same rule would work unchanged for a different asset universe or a materially different position limit, and penalty miscalibration remains one of the most commonly cited practical weaknesses of this style of quantum optimization formulation generally.

All quantum circuit execution in this study was performed on a noiseless simulator. No hardware noise, whether from imperfect gates, qubit decoherence, or measurement error, was included in any result reported above, so the confidence problem documented in Section 7.2 should be understood as a best-case outcome; on real, current quantum hardware, the share of valid and optimal measurements would very plausibly be worse, not better. Executing the same circuits on real quantum hardware, to measure exactly how much worse, is a natural and currently unexplored extension of this work.

The evaluation of each selected subset's quality assumes equal weighting across the chosen assets rather than a further continuous reallocation within the selected subset. This keeps the discrete and continuous problems cleanly separated for the purposes of this study, but a real portfolio manager would very likely re-run a continuous optimization restricted to the selected assets once they are chosen, which would improve on the equal-weight Sharpe ratios reported in Table 2; combining the discrete selection step with a subsequent continuous allocation step is a natural extension.

Finally, the eleven-asset universe examined here, while large enough to expose genuine and informative QAOA behavior, remains small enough that brute-force verification is itself entirely practical, completing in under ten milliseconds at every tested limit. The contribution of this study lies in the benchmarking methodology, not in the claim that quantum computation offers any presently realizable advantage at this particular scale; no such claim is made.

9. Conclusion

This paper examined, on a problem small enough to verify exactly, whether the quantum algorithm currently being piloted by financial institutions for cardinality-constrained portfolio selection can be trusted to deliver a reliable answer. The continuous allocation problem was solved exactly and instantly by classical convex optimization, for which no quantum alternative is needed or appropriate. The discrete selection problem, choosing the best fixed-size subset of assets, was solved using QAOA, benchmarked rigorously against both exact brute-force enumeration and classical simulated annealing. Across five tested position limits, QAOA's best-of-five-restarts measurement distribution was found to contain the true optimal portfolio in every case, a genuine and creditable result, but the algorithm did not concentrate confidence on that answer, and the share of measurements satisfying even the basic position limit varied considerably across limits without a clean pattern. The practical implication for financial institutions is specific: QAOA today demonstrates real algorithmic capability at small scale but does not yet provide a confidence signal a portfolio manager could act on without independent verification, a precise diagnosis of why some institutions continue to invest in this technology while others have scaled back, and a clearer target for what future work, on better optimizers, deeper circuits, and lower-noise hardware, would need to fix before quantum methods could plausibly take on cardinality-constrained portfolio selection in practice.

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